

PORTFOLIO PRESENTATION

ARTIFICIAL INTELLIGENCE

Osama Mohammed Essmat

AI & Computer Vision Engineer

Architecting next-generation **Autonomous AI Agents** and **RAG Systems** combining custom LLM orchestration with production-grade engineering to transform complex workflows into intelligent decision-making solutions.

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Current Role: ML Engineer Trainee & Team Lead, DEPI

About Me & Strategic Vision

Bridging theoretical machine learning with robust production software

Background & Philosophy

My journey into AI started with a fundamental fascination: *how can a machine "see" and truly understand the physical world as humans do?* That curiosity evolved into specialized expertise in **Computer Vision, Deep Learning, and LLM Orchestration**.

I strongly advocate that real engineering mastery comes from **building end-to-end applications** rather than theoretical reading alone. Every system I build is an opportunity to optimize performance, refine architectural scalability, and solve complex real-world challenges.

Core Expertise & Pillars

- **Generative AI & RAG:** LangChain, FAISS, Vector Databases, Qwen, Custom Prompt Engineering, and LoRA Fine-Tuning.
- **Computer Vision:** Real-time Object Detection (YOLOv8), OpenCV Matrix Processing, SIFT, and Harris Corner Feature Extraction.
- **Deep Learning Architecture:** CNNs, LSTMs, 1D-CNNs with PyTorch, TensorFlow, and Keras.
- **Deployment & Systems:** Streamlit, CustomTkinter, Python C++ Embedded Integrations (ESP32 / Wokwi).

Featured Projects: Flagship AI Applications

Intelligent systems leveraging RAG and custom multi-modal workflows

NutriMind AI

AI-POWERED NUTRITION ASSISTANT USING INBODY REPORTS & LLMS

An end-to-end health intelligent system that ingests InBody clinical scan PDFs, extracts key body composition metrics, and constructs a customized dynamic knowledge base.

- **PDF Parsing & Extraction:** Structured extraction of muscle, fat, and hydration metrics.
- **RAG Pipeline:** Vectorized retrieval over personalized medical/fitness data for contextual LLM answers.
- **Interactive Dashboard:** Real-time guidance and customized nutrition/workout plans.

RAG

LLMs

PyPDF

Vector DB

Streamlit

EduMind AI

COMPREHENSIVE AI-POWERED PERSONAL LEARNING PLATFORM

An all-in-one educational workspace that converts static course materials and lecture PDFs into dynamic, interactive learning assets.

- **AI Quiz & Flashcard Generator:** Automated extraction of key concepts into practice sets.
- **Lecture PDF Assistant:** Chat directly with course material using precise context grounding.
- **Analytics Dashboard:** Tracks student performance, weak topics, and active recall streak scores.

LangChain

Qwen

FAISS

NLP

Custom UI

Computer Vision & Advanced NLP Engineering

Real-time edge detection systems and hybrid semantic search architectures

Driver Drowsiness Detection REAL-TIME FACIAL MONITORING

Safety application monitoring driver alertness using continuous video stream analysis and micro-sleep detection.

- **Custom YOLOv8 Model:** Optimized facial keypoint mapping.
- **Thread-Safe Audio Alert:** Instant acoustic triggers via Pygame.mixer.
- **Custom UI:** Modern desktop GUI with CustomTkinter.

YOLOv8

OpenCV

CustomTkinter

Hybrid Keyword Extractor NLP WEB APPLICATION

System combining classical statistical features with semantic deep learning embeddings for robust keyword scoring.

- **Hybrid Architecture:** TF-IDF Vectorization combined with sentence-transformers.
- **Preprocessing:** NLTK and SpaCy linguistic filtering and tokenization.
- **Cosine Similarity:** Normalized semantic relevance mapping.

Transformers

SpaCy

Streamlit

Instant Language Translator MULTI-LINGUAL DESKTOP UTILITY

Desktop application providing instant multi-language translation using deep-translator API wrappers.

- **Multi-Engine Support:** Google Translate backend integration.
- **Clipboard Utility:** Instant 1-click clipboard translation copying.
- **Lightweight GUI:** Responsive layout built for high performance.

Python

deep-translator

Tkinter

LET'S BUILD THE FUTURE OF AI

Thank You!

Open for collaborations, AI engineering roles, and groundbreaking machine learning projects.

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